LAB 8 - Unbalanced Wheatstone Bridge
TECH 60 - Sec 14
F26

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Objective:
The objective of this lab is to understand the Wheatstone Bridge and the relationship between voltages and currents in an unbalanced Wheatstone Bridge circuit. We compare hand calculations, Multisim simulation, and a breadboard circuit, both unloaded and with a 10 kΩ load between points A and B.

Equipment/Materials:
Solderless Breadboard
Keysight EDU36311A, Triple Output, Programmable DC Power Supply
Keysight EDU34450A 5½ Digit Digital Multimeter
Multisim software running on PC
1500 Ω resistors (Quantity 1)
750 Ω resistors (Quantity 1)
470 Ω resistors (Quantity 1)
390 Ω resistors (Quantity 1)
10 kΩ resistors (Quantity 1)
1 kΩ potentiometer
Jumper wires

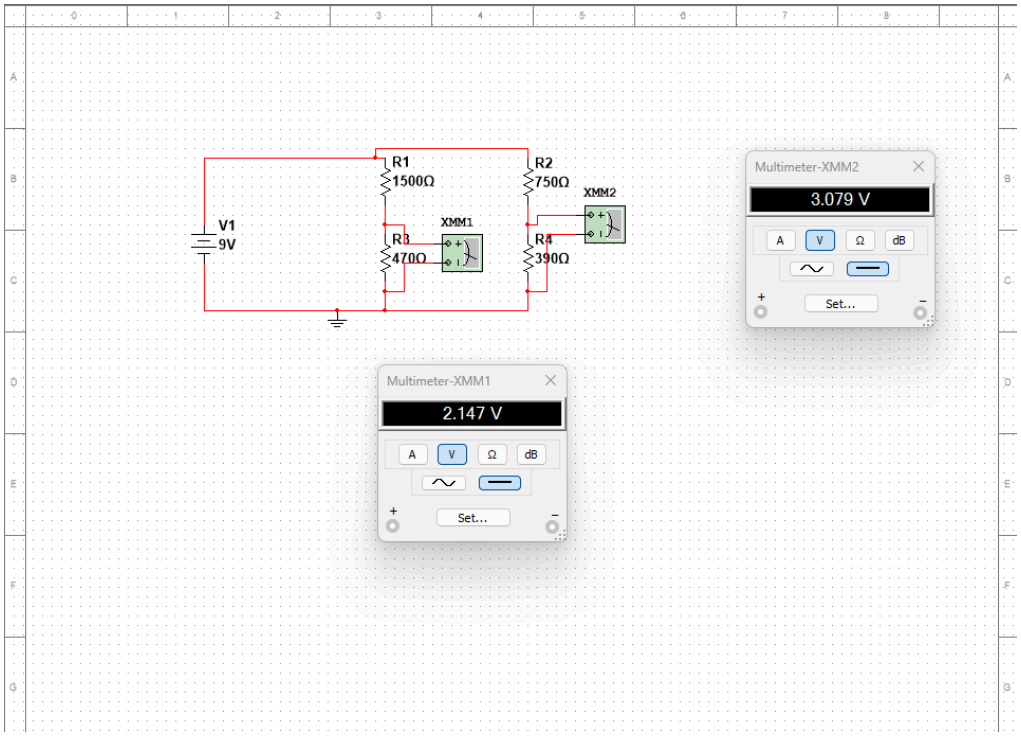
Part I. Hand Calculations
$V_s = 9$ V
$R_1 = 1500\ \Omega$
$R_2 = 750\ \Omega$
$R_3 = 470\ \Omega$
$R_4 = 390\ \Omega$

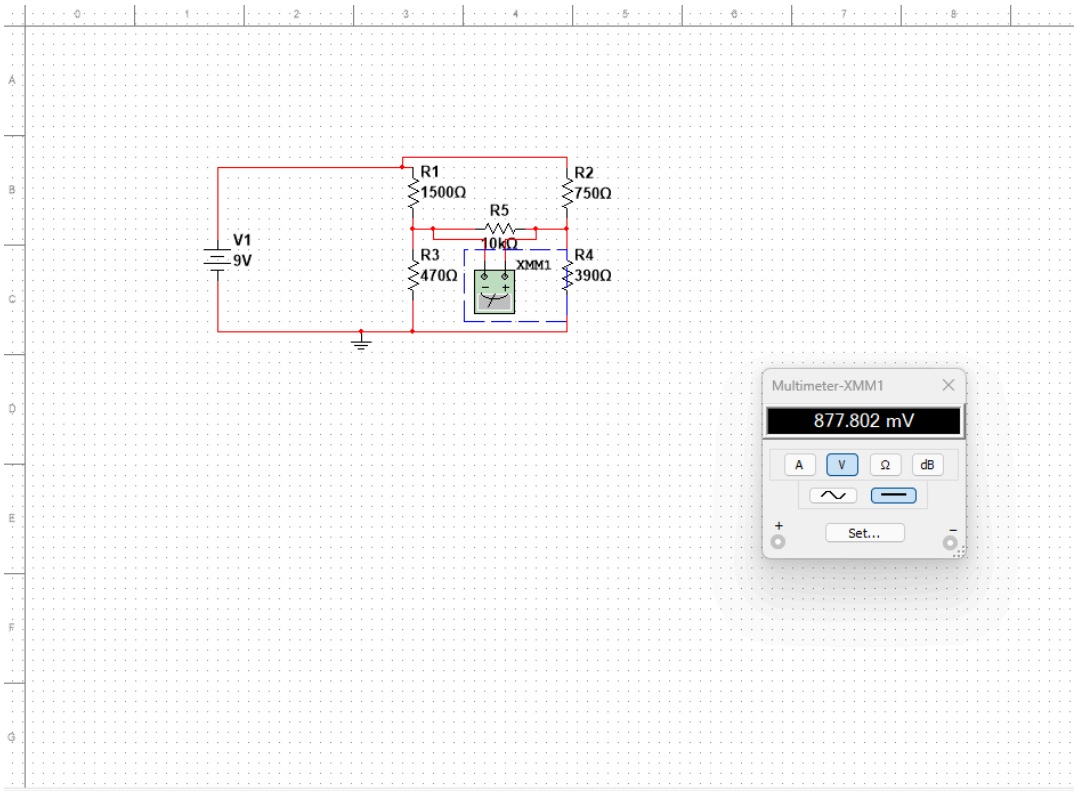
Voltage at A (voltage divider with R_1 and R_3)
$V_A = V_s \times R_3/(R_1 + R_3)$
$V_A = 9\ V \times 470\ \Omega/(1500\ \Omega + 470\ \Omega) = 4230/1970 = 2.147$ V

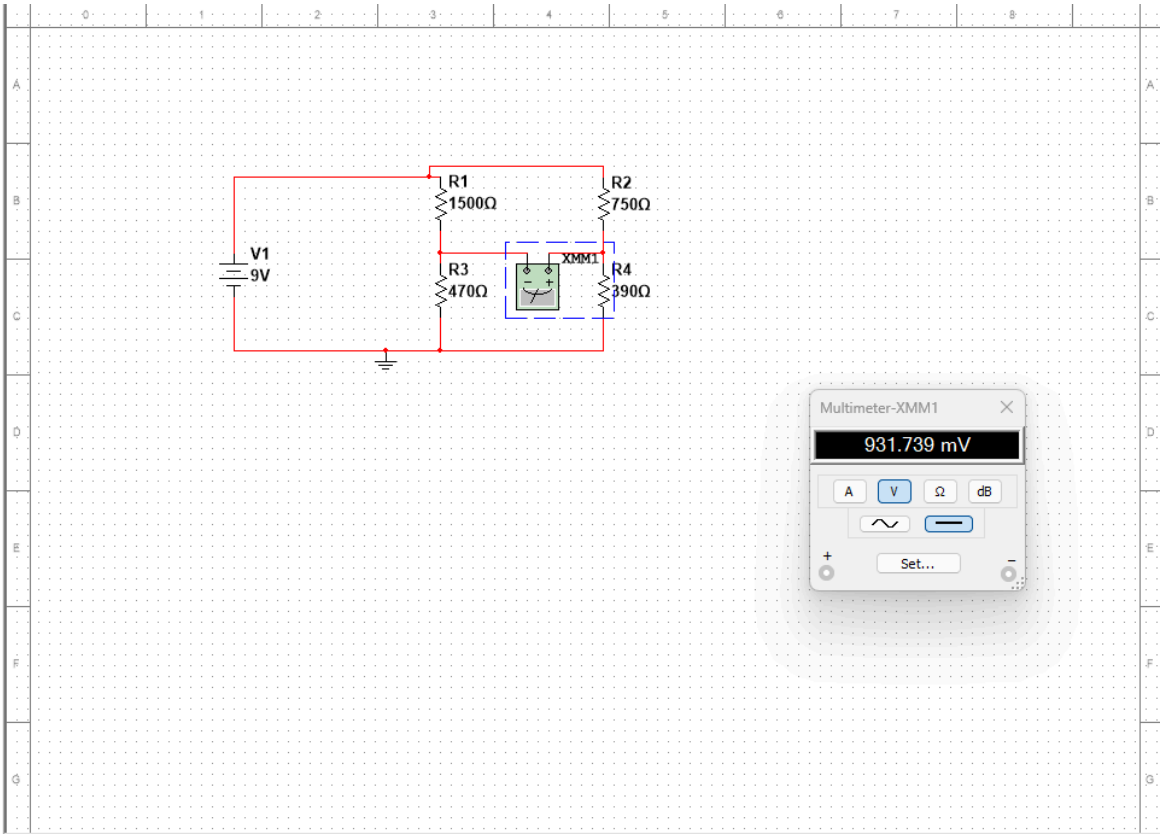
Voltage at B (voltage divider with R_2 and R_4)
$V_B = V_s \times R_4/(R_2 + R_4)$
$V_B = 9\ V \times 390\ \Omega/(750\ \Omega + 390\ \Omega) = 3510/1140 = 3.079$ V

Output voltage
$V_{OUT} = V_B - V_A$
$V_{OUT} = 3.079\ V - 2.147\ V = 0.932$ V

Part I


R1
1500Ω
R2
750Ω
XMM2
V1
9V
XMM1
R3
470Ω
R4
390Ω
Multimeter-XMM2
3.079 V
A
V
Ω
dB
Set...
Multimeter-XMM1
2.147 V
A
V
Ω
dB
Set...


R1
1500Ω
R2
750Ω
R5
10kΩ
V1
9V
R3
470Ω
XMM1
R4
390Ω
Multimeter-XMM1
877.802 mV
A
V
Ω
dB
Set...



I.

Circuit simulation using Multisim

Part III. Circuit construction on the breadboard

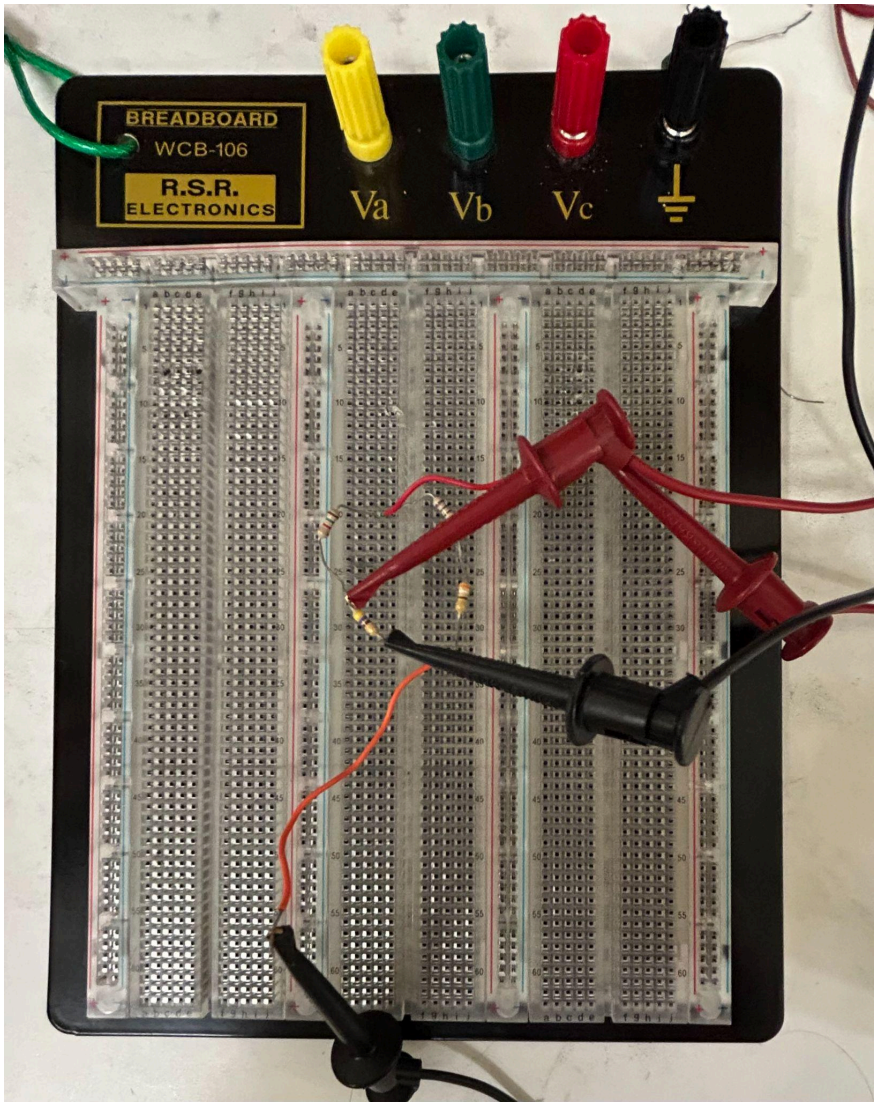


Figure 1. Unbalanced Wheatstone Bridge measuring V_A

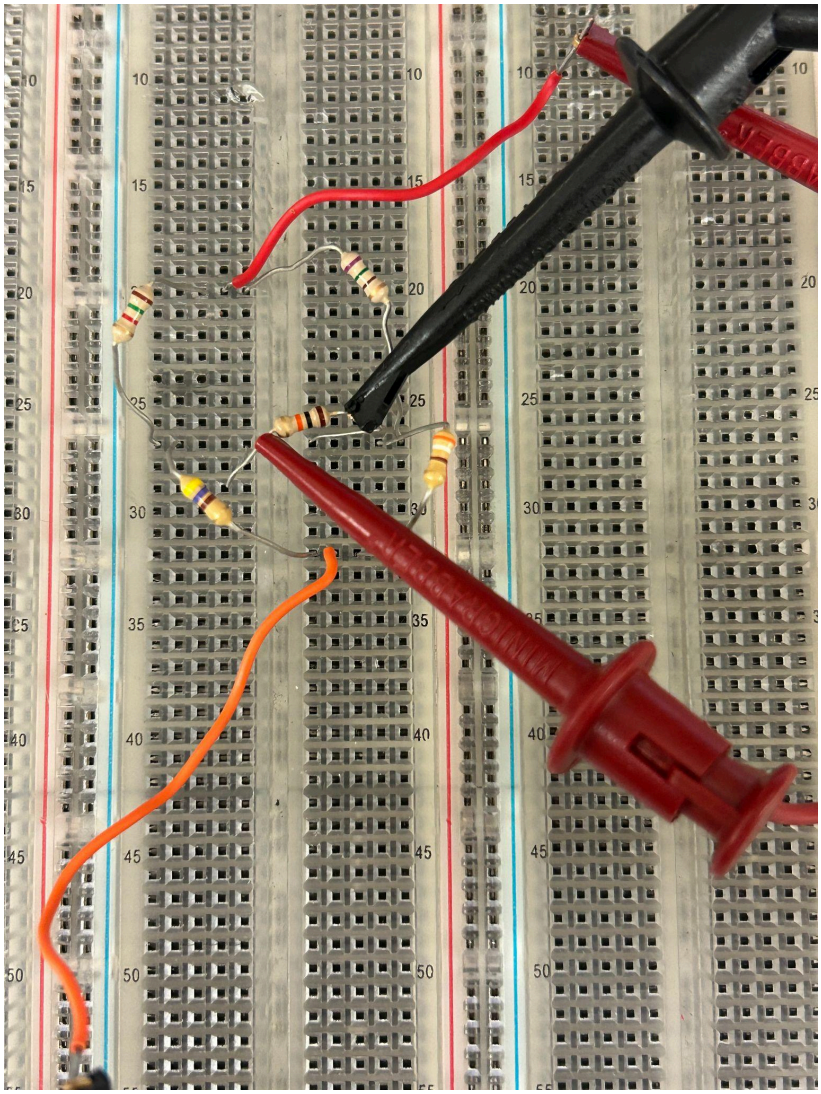


Figure 2. Unbalanced Wheatstone Bridge with load resistor connected

Part IV. Questions

1. Compare the measured resistor values to the expected values. If they are not exactly the same then explain briefly why they are not exactly equal.

The measured resistor values are slightly less than the expected values for all resistors in this lab. This is due to the fact that resistors are given tolerance rates to account for manufacturing imperfections. Also there are other conditions in the lab such as human error, breadboard resistance, wire resistance, DMM resistance, etc. that can alter the measurements.

2. Compare the measured and the Multisim values of voltages and explain any differences.

The measured voltage values are smaller than the expected values we found in Multisim. This is due to Multisim being simulated in perfect and ideal conditions. Once again there are conditions in the lab that alter the values slightly such as human error, resistance within all of the components, such as the jumper wires, breadboard, DMM, and voltage source, as well as air and other resistances. Our highest voltage error percentage was with the V_OUT measurement. The error percentage was higher than the rest due to the value being smaller and therefore there is more room for error ion terms of percentages.

3. From your experimental results have you gained a good understanding of Wheatstone Bridge circuit?

Yes, we gained a good understanding of the Wheatstone Bridge, including how it is drawn as a schematic and how to build it in Multisim and on a real breadboard. Because Multisim cannot rotate components diagonally, we drew the bridge as two series pairs of resistors (R1 and R3, R2 and R4) connected in parallel across the source, with open terminals at the midpoint of each pair (A and B) to measure VOUT. This is electrically the same circuit as the diamond-shaped Wheatstone Bridge. On the breadboard, we built the same layout.

Figure 3. Resistance Data

	R1	R2	R3	R4
Expected value	1500 Ω	750 Ω	470 Ω	390 Ω
Measured	1.487 kΩ	743 Ω	468 Ω	383 Ω
%Error Between Expected Value and Measured	0.87%	0.93%	0.43%	1.79%

Figure 4. Unbalanced Bridge Data

	VA	VB	VOUT	VOUT, Loaded
Hand Calculations	2.147 V	3.079 V	931.739 mV	N/A
Multisim	2.147 V	3.079 V	931.739 mV	877.802 mV
Breadboard	2.134 V	2.945 V	826 mV	861 mV
%Error Between Hand Calculations and Breadboard	0.61%	4.35%	11.35%	1.91%

Percent Error Calculations

%Error = |Measured − Expected| / Expected × 100%

Challenges:

One challenge in this lab was connecting the resistors correctly on the breadboard to create an unbalanced Wheatstone Bridge circuit. It was difficult to go through but eventually we solved it. Another challenge was measuring the voltage at points A and B accurately using the digital multimeter. We had to use the jumper wire cables but we were using them incorrectly so we asked for assistance. We also had to compare the measured resistor values with their expected values because they were not exactly the same. Small differences in resistance could affect the voltage readings and cause differences between the hand calculations, Multisim simulation, and breadboard measurements. We made sure that all the connections were correct and understanding these differences were important parts of completing the lab.

Conclusion:

In conclusion, this lab helped us understand how an unbalanced Wheatstone Bridge circuit works and how voltage changes depending on the resistor values. We used hand calculations, Multisim simulations, and a breadboard circuit to compare the results. The calculated voltages at points A and B were 2.147 V and 3.079 V, while the measured breadboard voltages were 2.134 V and 2.945 V. These results showed small differences, which could be caused by resistor tolerances and measurement errors. We also observed through Multisim that adding a 10 kΩ load decreased the output voltage. Overall, this lab improved our understanding of voltage dividers, resistor relationships, and how to analyze an unbalanced Wheatstone Bridge circuit.